

Multi Head Graph Attention for Drug Response Prediction

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Abstract— Precision medicine is based on curing diseases based on a patient's genetic profile, lifestyle, and environmental factors. This method improves clinical trial success rates and speed up drug regulatory approval. Predicting tumour vulnerability to specific anti-cancer therapy is critical for the successful implementation of precision medicine. Drug combinations have been shown to be very effective in cancer treatment to lower the drug resistance and improve the therapeutic effectiveness. The experiments carried out in all these therapeutic combinations have become expensive and time-consuming as a result of the increasing number of anti-cancer drugs. Large-scale drug response testing on cancer cell lines might help to understand the way drugs react with cancer cells. This study proposes a multi head graph attention network to perform drug response prediction.

Keywords: Multi head graph attention network, anti-cancer therapy, precision medicine, drugs resistance.

I. INTRODUCTION

Precision medicine is a rapidly growing field that aims to improve the effectiveness of medical treatment by tailoring it to the specific characteristics of individual patients. This approach is particularly promising in the field of oncology, where it has the potential to find patient subgroups with similar molecular profiles, lifestyles, and environmental factors with the aim of increasing the success rate of Phase II and III clinical trials. With oncology at the forefront of its development, the use of precision medicine has increased the number of applications in medicine and healthcare. Advances in high-throughput screening methods have enabled the profiling of cancer genomic signatures, providing insight into the molecular characteristics of tumors that can predict response to specific cancer drugs. This has led to improved rates of disease prognosis. Large-scale initiatives that concentrate on drug response in "artificial patients" (cell lines), such as GDSC, CCLE, and NCI60, have also been highly significant in the development of computational techniques for drug response prediction. Additionally, the DREAM challenge has been launched to further advance this field, with many research groups proposing new methods for drug sensitivity prediction.

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Genomics of Drug Sensitivity in Cancer and The Cancer Cell Line Encyclopedia and projects are working towards advancing personalized medicine by compiling and analyzing genomic and pharmacological data on human cancer cell lines. CCLE has collected data on 947 cell lines and 24 anti-cancer drugs, while GDSC has focused on 639 cell lines and 130 drugs. These databases seek to identify genetic biomarkers of drug sensitivity in cancer cells and take various forms on gene expression, DNA copy number, and oncomap mutations. Both studies make use of cell lines from several human cancers, include lung, breast, and kidney. Datasets from high-throughput screening have significantly vital in advance cancer and research care. These datasets have been used to investigate a wide range of topics, such as the discovery of new cancer drugs, understanding the underlying mechanisms of human cancer, and repurposing existing drugs. Many of these studies combine various types of data, including gene expression and treatment response, using machine learning methods and methods like multiple-kernel and multiple-task learning and ensemble learning. Using pharmacological and cell line omics data to predict drug response, deep learning is a sophisticated branch of machine learning that is well-suited to deal large amounts of high-dimensional and complex data. It has been demonstrated to be at least as effective as traditional machine learning techniques, if not more so, in a variety of tasks relevant to drug discovery. Deep learning models have been used in the field of drug response prediction to learn the genetic characteristics of cell lines or the structural characteristics of pharmaceuticals to predict cancer drug responsiveness. These models include artificial neural networks, CNN, and stacked deep auto-encoders. However, these models may not fully capture the structural information of drugs as they are often represented as strings. Graph neural networks are a technique that is utilized to analyze complex relationships present in molecular graphs which are often used to represent compounds in various fields like drug discovery. These networks have been demonstrated to be successful in predicting certain properties, such as drug-target binding affinities. However, their use in forecasting drug response has not yet been researched. Given the capacity of graph neural networks to model interactions within graph-structured data, it is an area that is worth considering for future research. When compared to more conventional machine learning techniques, deep learning methods frequently make predictions better, but the downside is that they lack interpretability. To address this problem, techniques such as saliency maps have been developed. These maps originally

were used to identify important features in image classification tasks, but now have been applied in various fields such as video surveillance and traffic detection. The saliency map method could be used in this study to investigate the relationship between genetic variables, such as aberration features, and drug response prediction.

II. RELATED WORK

Graph convolutional neural networks (GCNNs) have recently been found to be useful in predicting medication response. In this study, we compared the performance of GCNNs to more established machine learning methods like gradient boosting and decision trees. Results indicate that GCNNs performed better than these methods. The model's input data included genomic attributes of cell lines, such as abnormalities and copy number variations, as well as chemical information about pharmaceuticals in SMILES format that was collected from PubChem. We utilized RDKit, a free chemical informatics system, to generate a molecular graph that reflected interactions between atoms in the drug. A collection of atoms that each have a feature vector developed using the deep-chem atom feature design were used to illustrate the molecular graph. The feature vector comprises five different sorts of information about an atom: its symbol, its degree, which is determined by how many hydrogen atoms are bound to it and its neighbor, its total amount of hydrogen atoms, its implicit value, and whether or not it is aromatic. These attributes are employed to give a thorough description of the molecules with in drug molecule, which is crucial for determining the drug response. For each medicine in this research, a chemical graph was constructed by connecting atoms that are chemically bound together with edges. This produced a binary graph containing nodes that may be ascribed, each of which is described by an input vectors. The features of the medications were then extracted from this graph using various graph convolutional neural models, including GCN, GAT, GIN, and a combination GAT-GCN architecture. In order to compress the output of the graph neural network to 128 dimensions, it was then passed via a fully connected layer. A standard method in deep learning models, 1D convolution layer were utilized for the genomic features. One-dimensional convolutional layers were used to process the genetic characteristics. This method is employed because the mixture of genomic abbreviations enables 1-dimensional convolution with a large kernel to produce predictions. In order to enhance the model's capacity to extract feature representations from genomic information, 1D pooling was also used to reduce the amount of the input features. One-hot encoding was used to represent the genomic characteristics of cell lines. The latent properties of such datasets were learned using 1-dimensional convolutional network (CNN) layers. The output was then compressed into a 128-dimensional vector representing cell lines. Graph convolutional network models are used to improve representation learning by utilizing deep architectures, but they still have limitations in their shallow structure. For instance, GCN usually only uses two layers and increasing the number of layers may lead to a decrease in performance. This is due to GCN's simple propagation process that can make the representations of nodes too smooth for distinct and far apart nodes. This goes against the purpose of using deep models. While some solutions have been proposed, such as skip connection-based models,

finding a deep architecture that can adapt to different graph structures is still an open challenge. To overcome this problem,

III. PROPOSED METHODOLOGY

We have introduced Multi-head GAT, which uses different projection matrices for each unique type of graph structure. Convolutional neural network (CNNs) with long short-term memory (LSTM) units, for example are frequently utilized in drug design and prediction tasks instead of CNNs with fully linked networks. A CNN-LSTM model can be used to evaluate single omic data, extract relevant characteristics like patterns or correlations from the data, and then utilize those features to forecast a drug's efficacy or toxicity. One use of a CNN-LSTM model in single omic data is in the prediction of drug response in cancer cells. The model can be trained on genomic data from cancer cell lines, such as RNA expression levels or mutations, and used to predict the sensitivity of the cells to different drugs. This can aid in the selection of drugs for clinical trials or personalized treatment plans for cancer patients. Another use of a CNN-LSTM model in single omic data is in the prediction of drug-protein interactions. The model can be trained on proteomic data, such as protein expression levels or post-translational modifications, and used to predict the binding affinity of drugs to specific proteins. This can help with finding new pharmacological targets or repurposing current medications for different uses. Overall, CNN-LSTM models can be a powerful tool in single omic data analysis for drug discovery and prediction tasks. They can extract useful features from the data and make predictions about drug efficacy or toxicity, drug-protein interactions, or other relevant information that can be used to aid in drug development and personalized treatment plans.

A. Architecture diagram

The components of drug response prediction model is illustrated in the following figure 1. This architecture explains about the process of smile string and cell line features feed into the drug

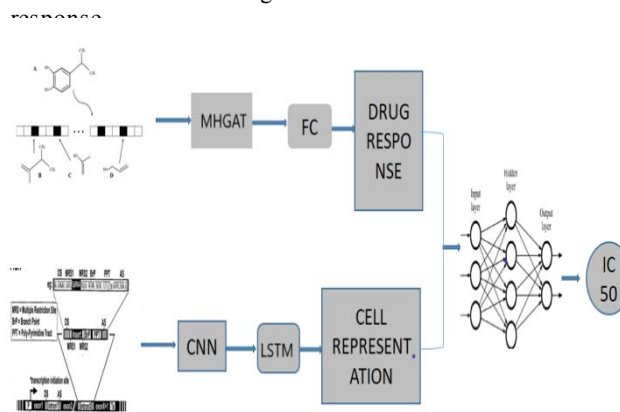


Fig. 1. MHGAT and CNN-LSTM is feed into the drug response prediction.

Fig. 1. represents graph structures of smile string are fed into multi head graph attention networks to encode the drug representation and, similarly, Cell line representation of gene mutations feed into the CNN with LSTM layers to

encode the representation, of cell line representation and both these are concatenated into the fully connected neural network to predict the IC50.

B. Multi head graph attention network

A form of neural network called the Multi-Head Graph Attention Net (MH-GAT) is made for activities based on graphs, such as node classification and graph classification. It is an extension of the standard Graph Attention Network (GAT) that utilizes multiple attention heads to weigh the importance of different features in the input graph. The key idea behind MH-GAT is to use multiple attention heads to learn different representations of the graph, and then concatenate them to obtain a final representation. Each attention head is responsible for learning a specific feature of the graph, such as global or local structures. By using multiple attention heads, the model can capture more complex relationships between nodes in the graph. The MH-GAT model can be trained using supervised or unsupervised learning, depending on the task at hand. In supervised learning, the model is developed using a labelled dataset with the intention of predicting that certain feature of a graph or its nodes. The model is trained using an unlabeled dataset in unsupervised learning with the aim of discovering the underlying graph structure. Numerous tasks, including node classifications, graph categorization, and community discovery, have been tackled using MH-GAT. It has been demonstrated to deliver cutting-edge performance on a number of benchmark datasets, including Cora, Citeseer, and Pubmed. A particular kind of neural network is the Multi-Head Graph Attention Net (MH-GAT) that can be used to predict drug response using single omic data. This network utilizes graph attention mechanisms to weigh the importance of different nodes in the graph, allowing it to learn and make predictions based on the underlying biological relationships represented by the graph. The graph proposed technique, which is typically implemented to use a dot-product attention function, serves as the foundation for the mathematical formulation for the MH-GAT algorithm. A set of node features are input the dot-product attention function, which then produces an attention coefficient for every of the graph's nodes. The importance of each node in the final forecast is measured using the attention coefficients. The specific mathematical expression for the dot-product attention function in the MH-GAT algorithm can be represented as follows:

$$\text{Atten}(Q_e, K_e, V) = \text{softmax} * V(Q_e K_e^T / \sqrt{d_k}) \quad (1)$$

The query, key, as well as value matrix are represented, respectively, by Q_e , K_e , and V_e in (1). The dimensions of the key is represented by d_k . Each attention head's dot-product attention factor is determined, and the resultant attention coefficients are then added together and fed into a feed-forward neuron to get the final prediction. This can be expressed numerically as:

$$\text{Pred} = \text{concat}(\text{atten}_1, \text{atten}_2 \dots \text{atten}_n) W_0 + b_0 \quad (2)$$

Where in (2) W_0 and b_0 stand for the biases and weights of a feed-forward neural net, respectively, and n stands for the amount of attention heads, Attention i indicates the attention coefficients determined by i th attention head. The network is usually trained on a data of drug response

measures and associated single omic data in order to train the MH-GAT. The discrepancy between the anticipated therapeutic response and the active drug reaction is known as the loss function. To reduce this loss, the network's weights and biases are adjusted using the gradient descent process. The following is a mathematical illustration of the loss function:

$$\text{Loss} = \|\text{Prediction} - \text{Actual Response}\|^2 \quad (3)$$

Where in (3) Prediction represents the predicted drug response and Actual Response represents the actual drug response. When using single omic data such as genomics, transcriptomics, or proteomics data, the Multi-Head Graph Attention Network (MH-GAT) algorithm first converts the omic data into a graph representation, where nodes represent biomolecules such as genes or proteins and edges represent biological interactions between them. It is important to note that the MH-GAT algorithm is a powerful tool that can be used to predict drug response using single omic data. It utilizes graph attention mechanisms and multiple heads to weigh the importance of different nodes in the graph and make predictions based on the underlying biological relationships represented by the graph. The algorithm is able to handle high-dimensional data, and the graph representation of the omic data can be easily modified to incorporate new information. The algorithm's adaptability enables it to be used with a variety of biological systems as medication classes. In order to enhance the predictions, the MH-GAT method can be integrated with additional methods like deep learning. Additionally, feature selection, dimensionality reduction, and other methods can be applied to enhance the algorithm's performance. It's crucial to remember that the omic data selection will rely on the precise study topic and the kind of medicine being examined. For example, if studying the effect of a drug on a specific gene, the genomics data would be used. In contrast, if studying the effect of a drug on protein expression, the proteomics data would be used. In summary, Multi-Head Graph Attention Network (MH-GAT) is a powerful algorithm that can be applied to predict drug response using single omic data. It utilizes graph attention mechanisms and multiple heads to weigh the importance of different nodes in the graph and make predictions based on the underlying biological relationships represented by the graph. The algorithm can be combined with other techniques, such as deep learning, and feature selection, dimensionality reduction to improve the predictions. The choice of the omic data will depend on the specific research question and the type of drug being studied. Graph DRP is a method that utilizes graph-based representations of biological interactions to predict drug response. This approach can be used to determine therapeutic efficacy in a variety of animals and cell types and is good at capturing the complicated interactions between medicines and biomolecules. However, it may have a high computational cost and have trouble handling huge or sparse graphs. MH-GAT is a neural network architecture that can be used for drug response prediction. It utilizes multiple attention heads to weigh the importance of different nodes in the graph, which allows it to make predictions based on the underlying biological relationships represented by the graph. This method can effectively learn and make predictions based on the underlying relationships represented in the graph.

However, it requires a large amount of labeled data to train the network effectively and also it may not be able to handle large and sparse graphs.

IV. IMPLEMENTATION

A. Datasets

To acquire a significant amount of data on how cancer cell lines respond to anti-cancer drugs, large drug sensitivity screening initiatives have been created. These studies, such as the Genomics of Drug Sensitivity in Cancer and also the Cancer Cell Line Encyclopedia, gather data on drug resistance in addition to the "omics" of cells, like gene expression. Data on gene expression, sometimes referred to as transcriptomic data, shows how much DNA is used to make RNA and how much protein is created by a cell. This information can provide insight into the level of activity of gene in particular cellular states, such as diseased or healthy. Genomic aberrations including mutation and copy number abnormalities in the genome are also included in omics data. Medicine response is determined by how well a drug slows the development of cancer cells. Drugs are administered to cell lines at various dosages, and the data obtained is typically expressed as an IC50 value (the amount needed to reduce the biological activity of the cells by 50%) or perhaps an AUC (area underneath the dose-response curve) number (a measure of the drug's overall effectiveness). While CCLE contains information on 504 cells examined with 24 medications, GDSC contains data on 1,074 cell lines evaluated against 250 pharmaceuticals, making it the largest collection of its type. After preprocessing, 223 medications and 948 cell lines were chosen for the analysis using the GDSC edition 6.0 data as the reference. 81.4% of the data, which was divided into pair of drug-cell lines, had response values, while 18.6% had none. In accordance with a prior study, the response values was normalized to use a range of 0 to 1. In the input step, cell lines were represented as a binary vector with 735 dimensions, indicating whether or not there were genomic abnormalities. Drugs were displayed using the standard SMILES format.

B. Data Preprocess

Removed missing or duplicate data, and ensured that the data is in a consistent format. To transformation of normalized the data, scaled the data, and/or transforming the data to make it suitable for analysis. To integrated multiple data sources, such as combining gene expression data with drug response data, to create a more complete dataset. to reduce the dimensionality of the data, for instance by feature selection, in order to make the data more comprehensible and improve the performance of the model. to separate the data into training and testing set in order to evaluate the effectiveness of the model.

C. Data Visualization

In the below Fig.2, the tissue type of all the lungs has the highest tissue type. Using the Value counts that make it visible, which is more data type

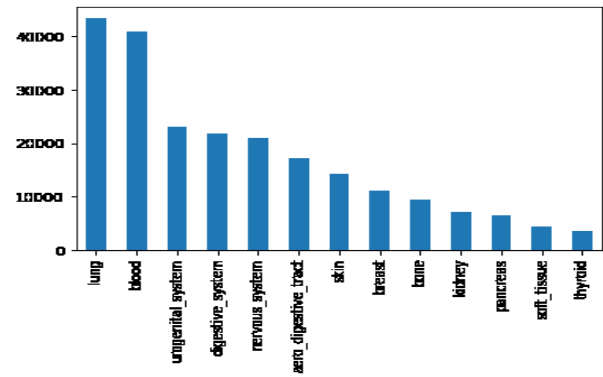


Fig.2. Tissue type in the cell line data set in single omic data

As below Fig.3 the descriptive analysis of the half-maximal inhibitory Concentration feature and it shows the normal distribution of the IC50.

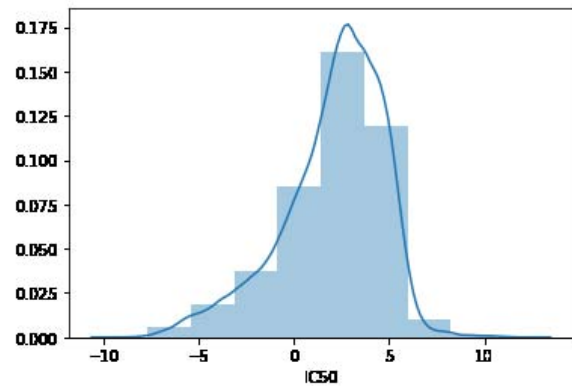


Fig.3. The distribution of IC50 values in single omic data.

V. EXPERIMENTATION

A. Multi-head graph layers

The graph layers in this model are responsible for processing the graph-structured data. The multi head convolution module is a graph attention convolutional layer that uses multiple attention heads to process the graph data. The first graph layer (gcn1) takes in the input graph data and applies the multi head convolution operation to it, while the second graph layer (gcn2) takes the output of the first graph layer and applies the same operation again. To reduce the output's dimensionality, a output of a second graphs layer is sent through a completely connected layer. A feature extraction of a graph data that can be applied to subsequent tasks like classification or regression is the output of the graph layers.

B. CNN-LSTM layers

The cell line layers in this model are responsible for processing the cell line data. A series of 1D convolutional, max pooling, and relu layers are employed in this section's opening to capture features from the cell line data. The output of this sequence is then sent into a layer of a recurrent neural network called an LSTM, which can interpret sequential input. To decrease the output's dimensionality, the output of a LSTM layer is sent via a fully connected layer. The cell line layers' final output is a

feature structure of the cell line data, which can be applied to subsequent tasks like classification or regression. To decrease the output's dimensionality, the output of a LSTM layer is sent via a fully connected layer. The cell line layers' final output is a feature structure of the cell line data, which can be applied to subsequent tasks like classification or regression.

C. Feed forward layers

The feed forward process in the above code refers to the forward pass of the data through the layers of the neural network. In the case of the MHGATNet class, the forward method takes in a single argument, "data", which contains the input for the network. The input is passed through several layers, including graph layers and cell line layers. The graph layers, which include the multi head graph convolution and linear layers, are used to process the graph-structured data. The multi head graph convolution layers are graph attention layers that perform convolution on the graph data, and the linear layers are used for fully connected layers. The cell line layers, which include the Conv1d, MaxPool1d, LSTM, and Linear layers, are used to process the cell line feature data. On the basis of cell line data, the Conv1d layer performs 1-dimensional convolution; the MaxPool1d layer performs max pooling; the LSTM layer performs long short-term memory; and the linear layer performs fully connected layers. Before the network generates a final output, the output of graphs layer as well as cell line layers were combined and passed through a number of fully connected layers. A activated sigmoid function is used to the output to give a likelihood value ranging from 0 and 1.

D. Training and Evaluation

The function starts by setting the model to training mode and defining a loss function, which is mean squared error loss. Then it initializes an empty list to store the average loss, then iterates over the batches in the train loader, where it loads the data and send it to the device. It then sets the gradients of the optimizer to zero, gets the output from the model, calculates the loss by comparing the output and the target, and performs the backward pass to calculate the gradients. Then it updates the model's parameters using the optimizer. The average loss is appended to the list. The train, validation, and test are loaded from processed files after determining a learning rate and the number of epochs. Then, the script makes the data ready for mini-batch processing using data loader, and sets the device to be used for training (either cuda or cpu) based on the cuda name and availability. A copy of the given model architecture is then created by the script, an Adam optimizer is established with the given learning rate, and the model is trained for the specified amount of epochs. The script also keeps track of the best model based on minimum mean squared error and Pearson correlation coefficient and saves the best model, test results, and figures of loss and Pearson correlation coefficient. The script appears to be designed to train and evaluate a deep learning model on a dataset called GDSC. The exact purpose of the model and how it is used to analyze the GDSC dataset is not clear from the code provided. However, it is likely that the model is being trained to perform some type of regression or classification task on the dataset.

It is possible that the modeling variable passed to the script is a function or class that defines the architecture of the model being trained. The Tested dataset class is used to load the train, validation, and test data from files and make them ready for mini-batch processing by py-torch. The script trains the model using the Adam optimizer, which is a popular optimization algorithm for deep learning. The training process is run for the number of epochs specified by the number of epoch variable, and the model's performance is evaluated on the validation set at regular intervals specified by the log interval variable. The script also keeps track of the best model based on the minimum mean squared error and pear-son correlation coefficient, and saves the best model and test results. Additionally, the script also plots the training and validation performance over time by generating figures of loss and Pearson correlation coefficient for the best model. In summary, this script is a training script for a deep learning model, it trains the model on GDSC dataset, saves the best model and test results, and plots the training and validation performance over time. However, the specific task or problem that the model is being trained to solve is not clear from the code provided. To train and evaluate a model on a dataset called GDSC. It allows the user to specify the model architecture, batch sizes, learning rate, number of epochs, log interval, and cuda name as command line arguments when running the script. To track of the best model based on the minimum mean squared error and Pearson correlation coefficient, and saves the best model and test results. Additionally, the script also plots the training and validation performance over time by generating figures of loss and Pearson correlation coefficient for the best mode allows the user to quickly and easily experiment with different model architectures, hyper-parameters, and settings to find the best configuration for their task. The script is designed to be flexible and modular, allowing different model architectures, datasets, and performance metrics to be easily swapped in and out.

TABLE I: EVALUATION OF CORREALTION COEFFICIENTS AND RMSE OF TRAINING

MODELS	Correlation coefficients	RMSE
MGHAT	0.9316	0.0234
tCNN	0.9160	0.0284
GAT-GCN	0.9308	0.0243
GAT	0.9270	0.0250
GCN	0.9216	0.0259

The TABLE I provides lists the correlation coefficients and root mean square errors (RMSE) for several models: MGHAT, tCNN, GAT-GCN, GAT, and GCN. A correlation coefficient's value of 1 indicates a high positive relationship, a value of -1 indicates a perfect negative correlation, as well as a value of 0 indicates no correlation. It assesses the direction and strength of the linear relationship between two variables. A lower value of the RMSE indicates a better fit as it measures the gap between expected and actual values. Based on the table, the MGHAT model has the highest correlation coefficient (0.9316) and the lowest RMSE (0.0234) among the models. This indicates that this model may have the strongest linear relationship with the variables

of interest and the smallest difference between predicted and actual values. The GCN model has the lowest correlation coefficient (0.9216) and the highest RMSE (0.0259) among the models.

E. Testing and Evaluation

In testing and evaluation, employs a machine learning model to forecast drug sensitivity for cancer cell lines, and then computes the gradients of the output with respect to the input characteristics (i.e., the mutations) to pinpoint the mutations that have the biggest influence on the model's predictions. Then, it calculates the gradients of the output with respect to the input features (i.e., the mutations) to find the mutations that have the biggest influence on the model's predictions. Loads a dictionary of mutations and converts it to an array. It then checks if the processed data file exists, and if not, it prints a message asking the user to run a script create new files to prepare the data. If the file does exist, the function loads the data using the tested dataset class and uses the specified model to make predictions on the data. The function then uses auto grad backward to calculate the output's gradients with reference to the input, and it chooses the most mutation with highest gradients. Finally, the function loads a list of cell lines from a file cell blind test and prints the top number of mutations and their gradients. It's a function that analyses the effect of specific mutations on the outcome of a drug sensitivity test. The function makes a prediction on a dataset of drug sensitivity for cancer cells using a trained machine learning model. It then computes the gradients of the output to reference to input attributes (i.e., the mutations) to pinpoint the mutants that have the biggest influence on the model's prediction models.

TABLE II: EVALUATION OF CORRELATION COEFFICIENTS AND RMSE OF TESTING

MODELS	Correlation coefficients	RMSE
MGHAT	0.8501	0.0365
tCNN	0.3490	0.0576
GAT-GCN	0.8402	0.0362
GAT	0.8321	0.0380
GCN	0.8399	0.0363

The Table II provides performance metrics for different machine learning models, specifically in the area of graph neural networks (GNNs). The name of the model being used appears in the first column. The correlation coefficients are listed in the second column, while the mean squared error is listed in the third (RMSE). The correlation coefficient, which runs from -1 to 1, is a measurement of the linear relationship between two variables. A perfect positive correlation is represented by 1, a perfect negative correlation by -1 and no connection by 0. The higher the correlation coefficient, the better the model is at predicting the target variable. The RMSE is a measurement of the discrepancy between expected and observed values. It is the mean squared error's square root. Lower values of RMSE indicate better performance of the model. From the data provided, it appears that the model with the highest correlation coefficient and the lowest RMSE is MGHAT, followed by GAT-GCN, GAT, GCN, and tCNN. The tCNN model has the lowest correlation coefficient and the highest RMSE,

indicating that it is the weakest performer among the models listed.

F. Saliency Map

Saliency in drug response prediction is a method used to identify the specific molecular features or genetic variations that are most important in determining a patient's response to a particular drug. This knowledge can be applied to find potential new therapeutic targets or develop individualised treatment plans in order to comprehend the underlying processes of pharmacological action. There are several different methods that can be used for saliency analysis, including feature importance measures and gradient-based methods. The relative significance of various molecular features or genomic in predicting medication response is assessed using feature importance metrics. Gradient-based methods, such as saliency maps, are used to identify the specific regions of a molecular feature or genetic variation that are most important for drug response prediction. Genetic, proteomics, and metabolomic data are only a few of the types of data that can be subjected to saliency analysis. It can also be combined with the other machine learning strategies, such deep learning, to boost the predictability of pharmacological responses. Overall, saliency analysis is an important tool in the field of personalized medicine as it can help to identify the specific molecular features or genetic variations that are most important in determining a patient's response to a particular drug, which can aid in the development of personalized treatment strategies.

TABLE III: THE TOP TEN GENOME ANOMALIES EXPECTED BY BORTEZOMIB TO JUST BE EFFECTIVE AGAINST NCI-H748, SW684, AND CELLS ARE LISTED IN ORDER OF SALIENCY

Anomaly	Saliency	Anomaly	Saliency
EWSR1-FLI1 mut	0.0144	cnaPANCAN84	0.0100
NSD1 mut	0.0060	cnaPANCAN247	0.0031
cnaPANCAN144	0.0039	cnaPANCAN13	0.0026
cnaPANCAN108	0.0035	cnaPANCAN384	0.0022
MAP3K4 mut	0.0027	cnaPANCAN383	0.0019
PCDH18 mut	0.0027	BMPR2 mut	0.0019
cnaPANCAN247	0.0027	cnaPANCAN22	0.0018
SRGAP3 mut	0.0025	cnaPANCAN211	0.0018
MLL mut	0.0023	cnaPANCAN143	0.0018
cnaPANCAN146	0.0021	cnaPANCAN377 0.0017	0.0017

The Table III provided appears to be a list of genome anomalies and their respective saliency scores for certain cell lines, with Bortezomib being expected to be effective against NCI-H748, SW684, and other cells. The top ten genome anomalies listed in order of saliency are EWSR1-FLI1 mut, NSD1 mut, cnaPANCAN84, cnaPANCAN144, cnaPANCAN108, cnaPANCAN13, MAP3K4 mut, PCDH18 mut, BMPR2 mut, and cnaPANCAN247. The saliency scores are also listed, with the highest being 0.0144 for EWSR1-FLI1 mut and the lowest being 0.0017 for cnaPANCAN377. The list appears to be from a study or research that aimed to identify specific genome anomalies

that may be targeted by Bortezomib, a proteasome inhibitor drug used in cancer treatment. The saliency scores indicate the level of importance or relevance of each anomaly to the effectiveness of Bortezomib in the cell lines mentioned. It is significant to emphasise that additional research is required to confirm the conclusions of this study, and it is possible that they do not uniformly apply to certain other types of cancer or cell lines. Also, The list suggests that EWSR1-FLI1 mutation, NSD1 mutation, and cnaPANCAN84 are the most salient genomic aberrations that might play a critical role in the sensitivity of cells to Bortezomib. Moreover, the study also indicates that these genomic aberrations are unique to certain cell lines and may not be present in others.

VI. CONCLUSION

Further, Multi head Graph attention network uses different testing scenarios to depict actual circumstances, include the blind study for untested medications, the visually - impaired line testing for untreated cells line (such as novel cell lines), and the mixture test with untreated drug-cell pairings (e.g., new drugs). The contrast the predictive accuracy between graph attention and the other two states-of-the-arts procedure, DeepDRs and MOLIs, that do not include clinical informations in those model, is also important with regard to the effectiveness of attention graphs of drug with omics datas of cell line in detecting drug responses. A CNN with LSTM can handle both high-dimensional data and sequential data, making it well-suited for drug response prediction that involves both image-based data (e.g. microscopy images) and time-series data (e.g. measurements of cell proliferation over time). Additionally, LSTMs have the ability to remember previous information and use it to make predictions, which can be useful in modeling the temporal dynamics of drug response. On the other hand, a CNN with fully connected layer is simpler and it has less computationally expensive. It can handle high-dimensional data well and make predictions based on features learned by the CNN. In summary, the choice between a CNN with LSTM and a CNN with a fully connected layer would depend on the specific dataset and problem at hand. If the dataset includes both high-dimensional and sequential data and the problem requires modeling temporal dynamics, then a CNN with LSTM may be more appropriate. However, if the dataset only includes high-dimensionality and the problem does not require modeling temporal dynamics, then a CNN with a fully connected layer may be sufficient.

Source code available in github:
<https://github.com/sivamamidi/Drug-reponse>

VII. FUTURE WORK

Multi-Head Graph Attention Networks (MH-GATs) and CNNs with LSTMs are both powerful architectures that could be used for drug response prediction in single omic data. However, they also have their own limitations. MH-GATs are relatively complex architectures that can be difficult to train and optimize. It may not be able to handle large graphs or datasets. It require graph-structured data, which may not be available for some drug response prediction problems. CNNs with LSTMs are designed to handle image-based data and time-series data, so they may not be well-suited for drug response prediction problems that don't involve these types of data. they are relatively computationally expensive, which may make them

impractical for large datasets or problems that require real-time predictions. They require a lot of data preprocessing, which can be time-consuming and difficult to accomplish. It's worth mentioning that both architectures have the limitation of overfitting, especially if the dataset is small, or the model is complex. Therefore, it's important to use techniques like regularization, cross-validation, and early stopping to prevent overfitting. In Future research should, however, implement a k-fold cross-validation system to improve the accuracy of the assess when evaluating the models' performances in making predictions. Last but not least, by learning drug properties from their graphs and using numerous genomic data of cell line, the predictions of pharmaceutical combination/drug synergy may be enhanced.

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